ADAPTIVE CONTINUOUS AUTHENTICATION USING BEHAVIORAL BIOMETRICS

Submitted By: Varada Anup
Roll no: MUT24MCA2058
Project Guide: Asst. Prof. Jiss Kuruvilla

PROBLEM STATEMENT

- Traditional authentication (passwords, OTPs, MFA) is static, verified only at login.
- Password reuse and phishing attacks lead to high account compromise rates.
- Existing continuous authentication systems often :
  - Are mobile-focused (not web-based)
  - Lack adaptability to user behavior changes
  - Don't integrate multi-modal behavioral patterns with context
- **Problem Statement**: Need a web-based continuous authentication system that dynamically verifies users based on behavioral biometrics, adapts to behavioral drift, and minimizes friction for legitimate users.

LITERATURE REVIEW

Title	Author(s)	Year	Source	Relevance
Behavioral Biometrics-Powered Continuous Authentication for Zero-Trust Remote Work Environments	Suleiman S. Abba et al.	2025	Asian Journal of Research in Computer Science	Real-time continuous authentication with ensemble ML + context
Continuous Authentication in the Digital Age: An Analysis of Reinforcement Learning and Behavioral Biometrics	Priya Bansal & Abdelkader Ouda	2024	Computers (MDPI)	RL-based keystroke continuous authentication model
Machine learning-based novel continuous authentication system using soft keyboard typing behavior and motion sensor data	Ensar Arif Sağbaş & Serkan Ballı	2024	Neural Computing and Applications	Multi-modal behavioral authentication using typing + sensors

OBJECTIVES

1. Develop a web-based continuous authentication system using behavioral biometrics.
2. Collect and analyze keystroke, mouse movement, and interaction patterns.
3. Build adaptive ML/DL models for anomaly detection in user behavior.
4. Enable real-time verification, alerting, and dashboard visualization of user authentication status.
5. Allow the system to adapt to behavioral drift, including mood or situational changes.
6. Provide a demonstrable prototype for live testing with multiple users.

TOOLS / PLATFORM

- Programming & Web Development:
  - Frontend: HTML, CSS, JavaScript
  - Backend: Node.js / Python Flask / Django
- Data Collection:
  - JavaScript listeners for keystroke/mouse events on the web page
- Machine Learning / DL:
  - Python libraries: scikit-learn, TensorFlow, PyTorch
  - Models: Autoencoder, Isolation Forest
- Visualization / Dashboard:
  - Chart.js / Plotly / React dashboard components
- Database:
  - MySQL / MongoDB to store user profiles and session data

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THE RISE OF AI-POWERED CYBER CRIME: DEEPFAKES, VOICE CLONING, AND AUTOMATED PHISHING ATTACKS

Submitted By: Varada Anup
Roll no: MUT24MCA2058
Project Guide: Asst. Prof. Jiss Kuruvilla

RELEVANCE OF THE TOPIC

- Rapid growth of AI has enabled sophisticated cyber crimes
- Deepfakes and voice cloning enable realistic impersonation
- Automated phishing increases attack scale and success rate
- Threatens digital trust, privacy, and financial security
- Highly relevant in the era of generative AI and digital communication

SCOPE AND SIGNIFICANCE

- Covers AI-driven cyber threats affecting individuals and organizations
- Applicable to banking, governance, social media, healthcare, and enterprises
- Highlights emerging risks beyond traditional cyber attacks
- Helps in understanding the need for advanced detection and regulation
- Supports awareness, prevention, and future cybersecurity research

OUTLINE

- Overview of AI-Powered Cyber Crime
- Evolution of Cyber Attacks Using AI
- Deepfake Technology and Criminal Misuse
- Voice Cloning Based Fraud Attacks
- Automated Phishing Using Generative AI
- Real-World Impact and Case Examples
- Detection, Prevention, and Mitigation Techniques
- Legal and Ethical Challenges
- Future Trends in AI-Driven Cyber Crime

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3. Yalçın, N., & Lale, B. (2025). Types of cyber-attacks with using voice. Journal of Scientific Reports-A, (061), 137-165.
4. Blake, H. AI-Powered Cyber Extortion: How Artificial Intelligence is Transforming Ransomware Attacks.

THANK YOU